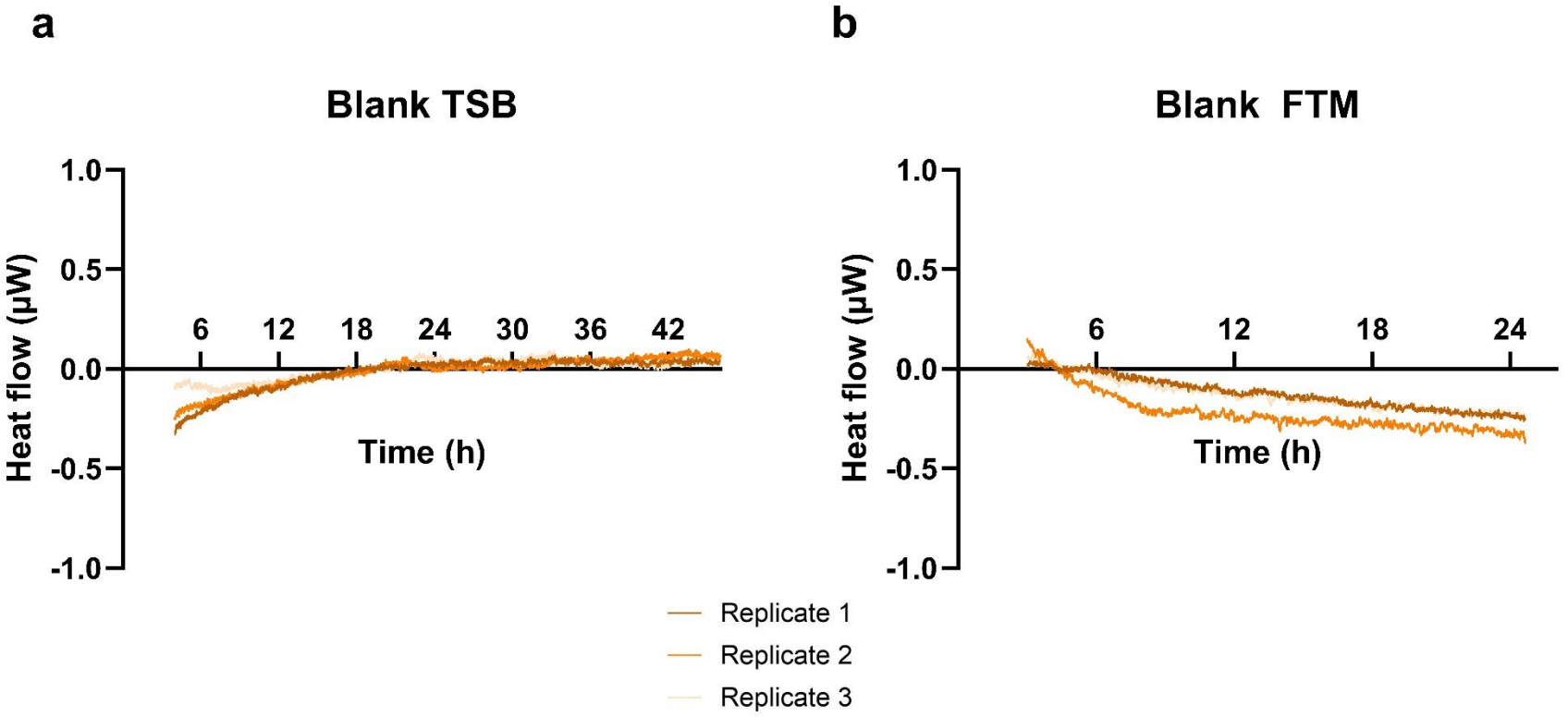
Online Resource 1 - Table Overview of all tested organisms with organism type and plate counts for 50 µL of suspension (mean ± standard deviation; n = 6).

		Plate counts (CFU)	
Organism	**Organism type**	**Aim 100 CFU**	**Aim 5 CFU**
***Acidovorax delafieldii* LPM-S1**	Gram-negative	30 ± 15	3 ± 2
***Aspergillus brasiliensis* ATCC 16404**	Mold	45 ± 22	3 ± 1
***Aspergillus versicolor* LPM-S2**	Mold	39 ± 10	3 ± 2
***Bacillus spizizenii* ATCC 6633**	Gram-positive	40 ± 12	2 ± 1
***Bacillus subtilis* LPM-S3**	Gram-positive	65 ± 18	2 ± 1
***Burkholderia contaminans* LPM-S4**	Gram-negative	65 ± 6	3 ± 2
***Candida albicans* ATCC 10231**	Yeast	52 ± 16	4 ± 2
***Clostridium sporogenes* ATCC 19404**	Gram-positive	37 ± 17	3 ± 2
***Cutibacterium acnes* LMG 16711**	Gram-positive	181 ± 49	11 ± 6
***Micrococcus luteus* LPM-S5**	Gram-positive	120 ± 24	7 ± 1
***Penicillium chrysogenum* LPM-S6**	Mold	55 ± 5	4 ± 2
***Pseudomonas paraeruginosa* ATCC 9027**	Gram-negative	72 ± 38	4 ± 2
***Staphylococcus aureus* ATCC 6538**	Gram-positive	89 ± 17	6 ± 3
***Staphylococcus epidermidis* LPM-S7**	Gram-positive	39 ± 15	3 ± 2
***Staphylococcus hominis* LPM-S8**	Gram-positive	36 ± 5	3 ± 2
***Streptococcus pyogenes* LMG 14237**	Gram-positive	22 ± 9	2 ± 1

Abbreviations: CFU, colony-forming unit



Online Resource 2 – Fig. A1 Thermograms of uninoculated samples of (a) TSB and (b) FTM, both in volumes of 400 µL. Short term noise and baseline drift equal 0.019 µW and 0.00069 µW/h for TSB and 0.041 µW and - 0.014 µW/h for FTM.

Online Resource 3 – Table Repeatability assessment of the calScreener+ sterility testing method. Results are based on three biological replicates, each consisting of two technical replicates (N = 6). Detection rate indicates the number of positive replicates (n) out of all tested vials. Mean relative standard deviation (%RSD) values, calculated based on the time to detection between technical replicates, are reported per organism, together with the range across biological replicates.

	Inoculum size			
	100 CFU		**5 CFU**	
Organism	**Mean %RSD (range)**	**Detection rate (n/6)**	**Mean %RSD (range)**	**Detection rate (n/6)**
***Acidovorax delafieldii* LPM-S1**	2.71% (1.03 – 4.58%)	6/6	12.04% (2.55 – 21.54%)[a]	5/6
***Aspergillus brasiliensis* ATCC 16404**	3.35% (2.03 – 5.89%)	6/6	23.75% (N/A)[b]	4/6
***Aspergillus versicolor* LPM-S2**	1.61% (0.24 – 2.32%)	6/6	8.34% (6.29 – 10.39%)[a]	5/6
***Bacillus spizizenii* ATCC 6633**	0.63% (0.39 – 0.97%)	6/6	N/A[c]	2/6
***Bacillus subtilis* LPM-S3**	1.35% (0.38 – 2.56%)	6/6	N/A[c]	2/6
***Burkholderia contaminans* LPM-S4**	0.60% (0.30 – 1.97%)	6/6	1.43% (1.08 – 1.77%)[a]	5/6
***Candida albicans* ATCC 10231**	0.55% (0.21 – 1.14%)	6/6	0.21% (0.18 – 0.23%)[a]	5/6
***Clostridium sporogenes* ATCC 19404**	19.54% (6.21 – 32.86%)	5/6	3.26% (2.02 – 4.55%)	6/6
***Cutibacterium acnes* LMG 16711**	2.40% (0.23 – 5.34%)	6/6	4.55% (0.69 – 9.43%)	6/6
***Micrococcus luteus* LPM-S5**	1.32% (0.17 – 2.47%)[a]	5/6	24.23% (N/A)[b]	3/6
***Penicillium chrysogenum* LPM-S6**	1.65% (0.42 – 2.76%)	6/6	3.50% (0.27 – 9.69%)	6/6
***Pseudomonas paraeruginosa* ATCC 9027**	9.95% (2.98 – 21.74%)	6/6	3.03% (2.46 – 3.99%)	6/6
***Staphylococcus aureus* ATCC 6538**	3.79% (0.43 – 6.64%)	6/6	4.26% (1.12 – 9.78%)	6/6
***Staphylococcus epidermidis* LPM-S7**	4.31% (1.20 – 7.24%)	6/6	7.26% (4.29 – 11.61%)	6/6
***Staphylococcus hominis* LPM-S8**	4.61% (3.17 – 6.48%)	6/6	42.32% (15.96 – 68.67%)[d]	4/6
***Streptococcus pyogenes* LMG 14237**	1.04% (0.12 – 2.85%)	6/6	5.10% (0.59 – 5.64%)	6/6
Overall	3.71%		10.24%	

Abbreviations: %RSD, relative standard deviation; CFU, colony-forming unit; N/A, not applicable

[a] Mean %RSD calculation based on two biological replicates, as for the remaining biological replicate, only one of the two technical replicates was detected

[b] Value equals the %RSD of a single biological replicate. Averaging was not possible because the remaining replicates were either single-positive or not detected

[c] Mean %RSD could not be calculated due to insufficient data: one biological replicate remained undetected; for the other two biological replicates, only one technical replicate was detected

[d] Mean %RSD calculated based on two biological replicates, as one biological replicate remained completely undetected

Online Resource 4 – Table Ruggedness assessment of the calScreener+ sterility testing method. Relative standard deviation (%RSD) values are calculated based on the time to detection between biological replicates (n = 3).

	Inoculum size	
	100 CFU	5 CFU
Organism	%RSD	%RSD
***Acidovorax delafieldii* LPM-S1**	7.25%	7.68%
***Aspergillus brasiliensis* ATCC 16404**	3.16%	11.17%
***Aspergillus versicolor* LPM-S2**	4.54%	8.99%
***Bacillus spizizenii* ATCC 6633**	1.96%	1.67%[a]
***Bacillus subtilis* LPM-S3**	8.07%	9.02%[a]
***Burkholderia contaminans* LPM-S4**	0.15%	1.33%
***Candida albicans* ATCC 10231**	12.98%	26.09%
***Clostridium sporogenes* ATCC 19404**	9.38%	1.73%
***Cutibacterium acnes* LMG 16711**	2.80%	8.03%
***Micrococcus luteus* LPM-S5**	1.69%	5.99%[a]
***Penicillium chrysogenum* LPM-S6**	2.34%	5.21%
***Pseudomonas paraeruginosa* ATCC 9027**	11.79%	10.01%
***Staphylococcus aureus* ATCC 6538**	3.27%	6.23%
***Staphylococcus epidermidis* LPM-S7**	1.89%	7.03%
***Staphylococcus hominis* LPM-S8**	4.66%	43.15%[a]
***Streptococcus pyogenes* LMG 14237**	2.41%	10.64%
Overall mean	4.89%	10.95%

[a] %RSD calculation based on two biological replicates, as one biological replicate remained undetected